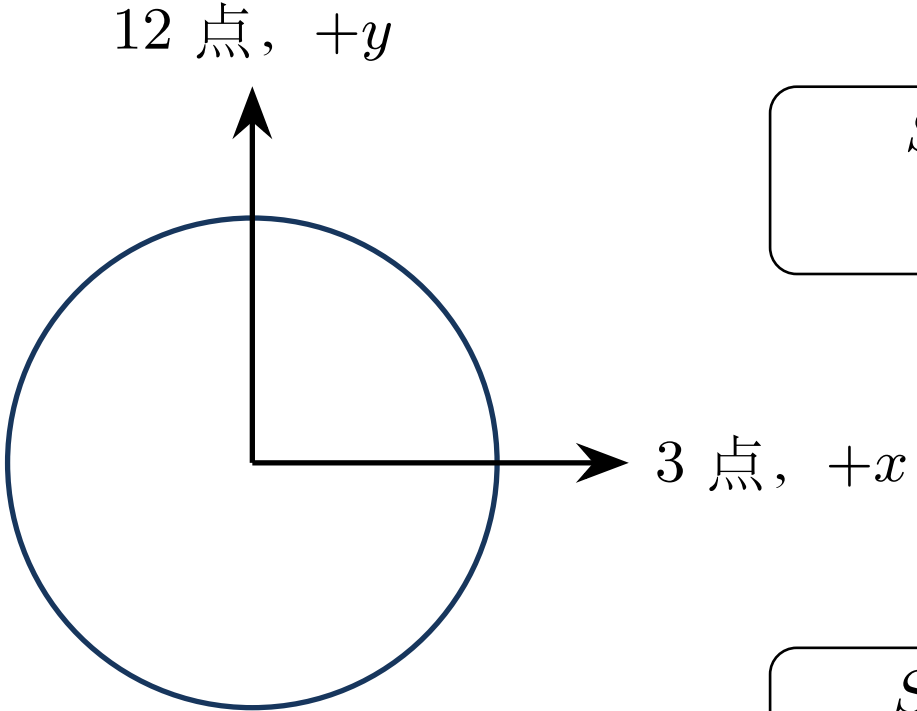
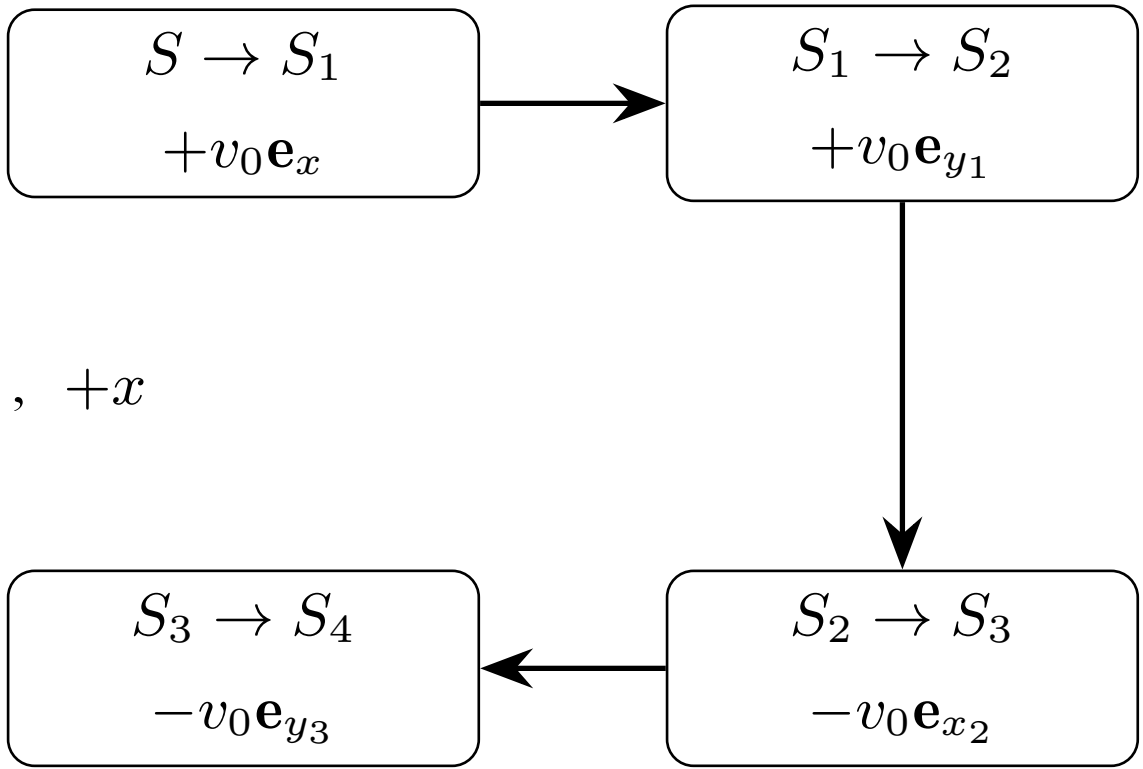
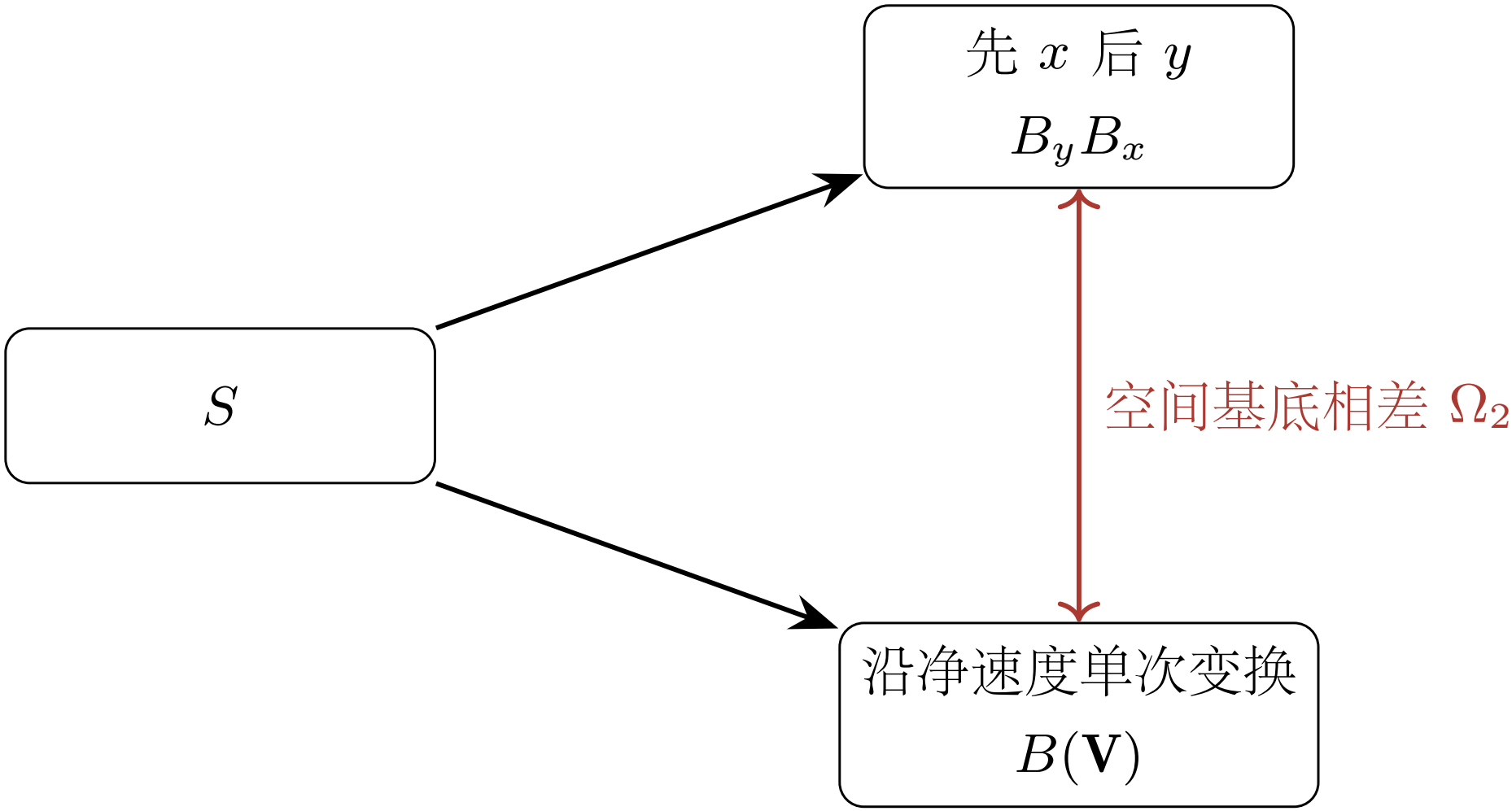
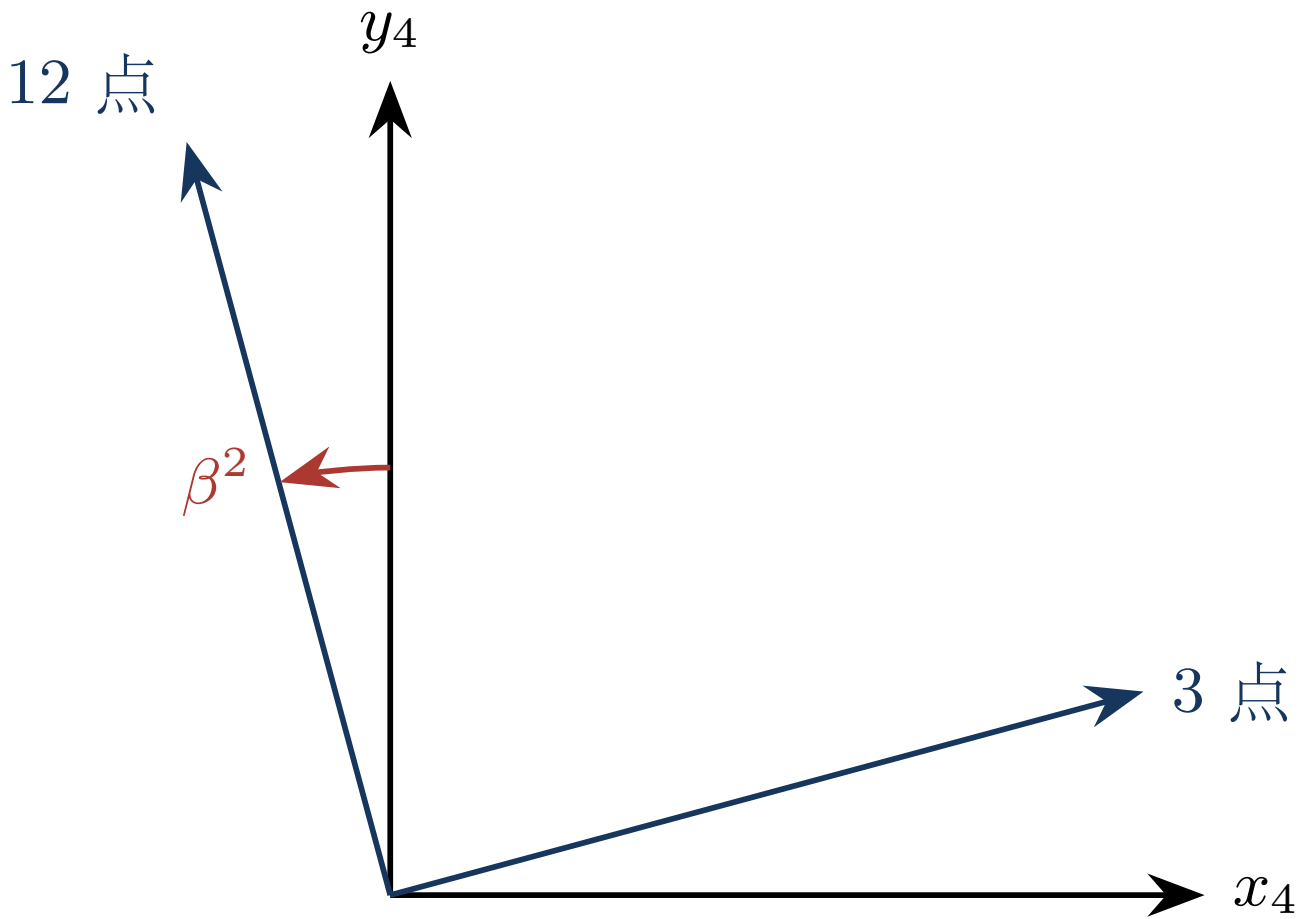

12 点，$+y$
3 点，$+x$
钟静止于 S


$S \to S_1$
$+v_0\mathbf{e}_x$
$S_1 \to S_2$
$+v_0\mathbf{e}_{y_1}$
$S_2 \to S_3$
$-v_0\mathbf{e}_{x_2}$
$S_3 \to S_4$
$-v_0\mathbf{e}_{y_3}$



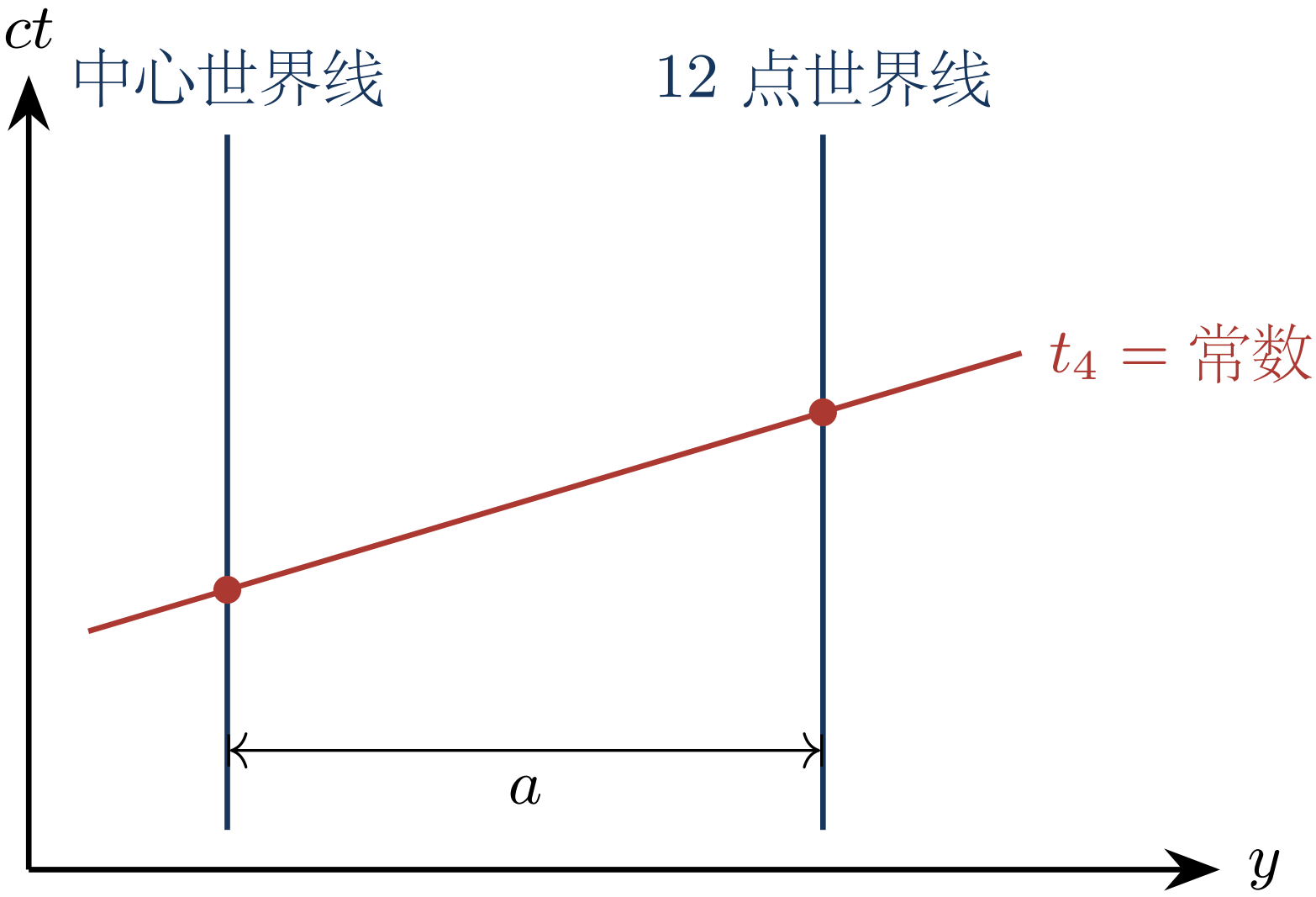
$\Omega_2 \simeq \beta^2/2$；坐标分量与坐标轴转向相反



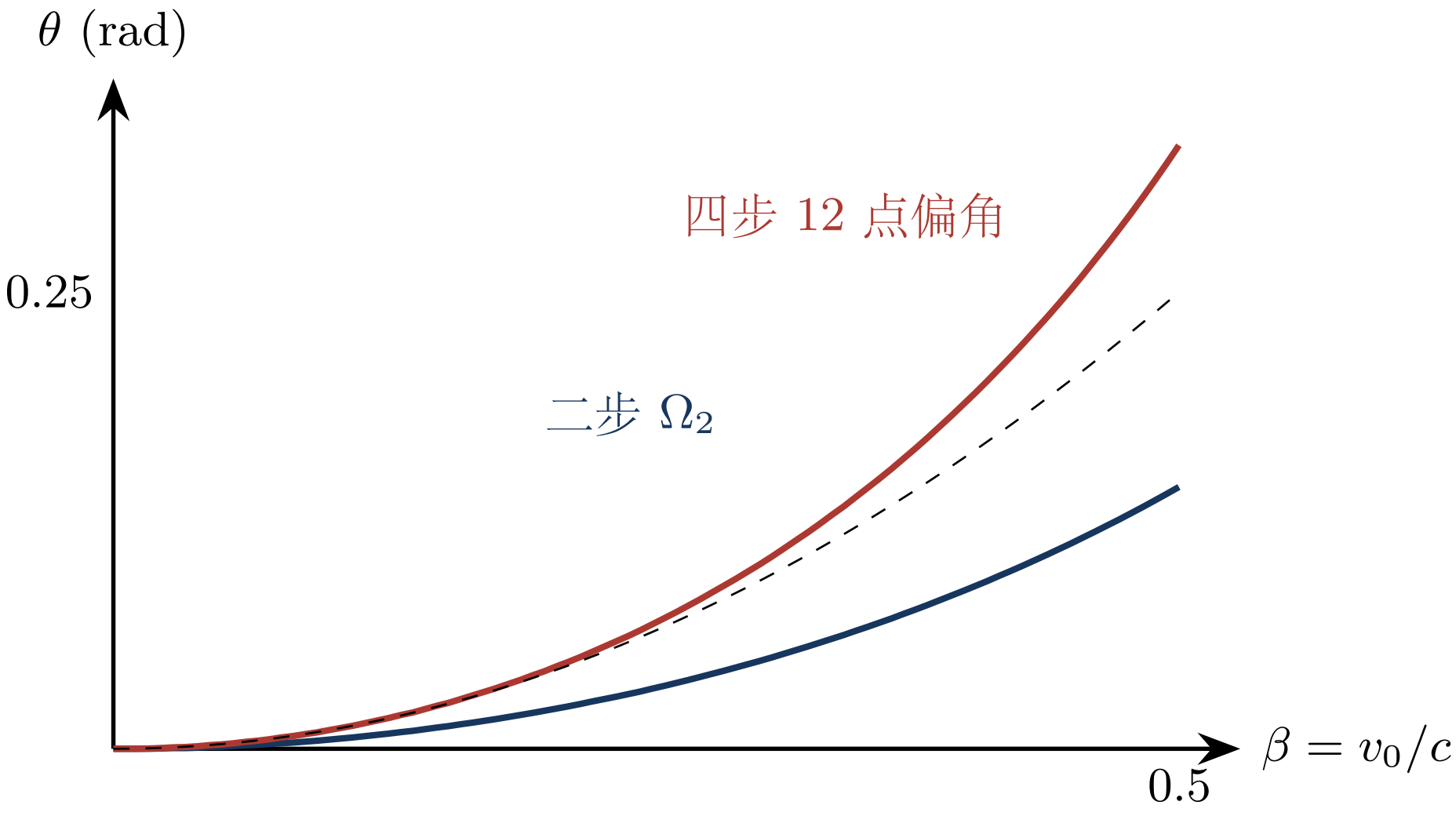
角度示意放大

转动 $O(\beta^2)$

残余速度 $O(\beta^3)$



同时条件 $M_{00}\Delta ct + M_{02}a = 0$



虚线：四步低速近似 β^2